

Start to Code with C++

ELEC1006: ENGINEERING COMPUTING

Header file

- A header file contains function declarations to be used in your code.
 - Header file has .h file extension.
- Called at the start of your code using `#include<file name>`.
 - Example:
`#include <iostream>`

Comments

- A comment is an annotation that adds more information to a specific line of code.
- It is not processed by the compiler and can be generated by either typing `// comment 1` or `/* comment 2 */`.
 - Use `// additional info` for single line annotation.
 - Use `/* additional info */` for multiple line annotation.
 - Example:

```
#include <iostream>    /* Function declarations in  
                        iostream.h can now be  
                        called in your code.*/
```

Namespace

- To call any function declaration from iostream.h header file, the `std::` prefix has to be used.

- Example:

```
std::cout << "Hello World\n";
```

- To shorten the code usage above to just `cout`, use namespace operator.

- Example:

```
#include <iostream>
```

```
using namespace std;
```

Function

- Your code must contain a main() function.
- When your program executes, the behaviour of your program is defined by the contents of main().
- It must be declared using a return-type identifier to tell the operating system (OS):
 - to inform the OS when the program ends so that the OS knows that extra computing resources (e.g. memory) are available for other programs.
- Example:

```
#include <iostream>
using namespace std;
int main()
{
    return 0;    // Program does nothing
}
```

Your 1st Program

- Use **cout** to display alpha-numeric characters (alphabets and numerals).

- Example:

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    cout << "Hello World";
```

```
    return 0;
```

```
}
```

```
// Display string
```

```
// Program exits
```

More info on C++

- [1] Learn Cpp.com: Tutorials to help you master C++ and object-oriented programming.
<https://www.learncpp.com/>